MGMTMSA 408 – Operations Analytics

Homework 4 – Network Flows and Traveling Salesman Problem (Question Sheet)

1 Papaya Republic Supply Chain

Papaya Republic is a new apparel retailer that specializes in clothing that "blends timeless Americana with modern, minimalist aesthetics". Papaya Republic is interested in managing its supply chain operations. In the firm's supply chain, a unit of a product corresponds to a container, which contains multiple garment.

Papaya Republic's supply chain network is organized in several layers:

Layer A consists of the manufacturing centers, which hold a certain number of container units of apparel in inventory. Layer B consists of warehouses, which also hold some number of container units in inventory. Layer C consists of the distribution centers. Finally, layer D consists of the retail centers, each of which is associated with some amount of demand in container units.

Papaya Republic is interested in understanding how its products should be transported across the supply chain network to meet demand at the lowest possible cost, and conducting some what-if scenarios to understand (1) what types of expansions to the supply chain network would most reduce cost, and (2) what types of disruptions to the supply chain network would be most detrimental to the supply chain cost.

To assist with these high level aims, you are provided with two data files. The file `MCNF_nodes.csv` is a CSV file containing information about each node in each layer in the supply chain network. Each row corresponds to a single node in the supply chain network. The columns of this file are described in the table below:

Column	Description
`NodeID`	Number to indicate the node; unique to each node.
`Layer`	String with single character representing layer of supply chain network that node is in; one of A, B, C, D.
`NodeName`	String with name of node (e.g., A4, which indicates fourth node in layer A). (First character will match `Layer`.)
`Inventory`	Number of container units already available at node.
`Demand`	Number of container units demanded at this node.
`BackorderCost`	Unit cost for each container unit that final supply is short of demand. (Only non-zero for nodes in layer D.)

Note that `Demand` and `BackorderCost` are only non-zero for the nodes in the final layer, which correspond to retail locations. Note also that `Inventory` is only non-zero for the nodes in layers A, B and C.

The file `MCNF_edges.csv` is a CSV file containing information about the edges of the supply chain network, along which container units can be moved from one node to another. Each row in this file corresponds to one edge in the supply chain network. The columns of the this file are described in the table below:

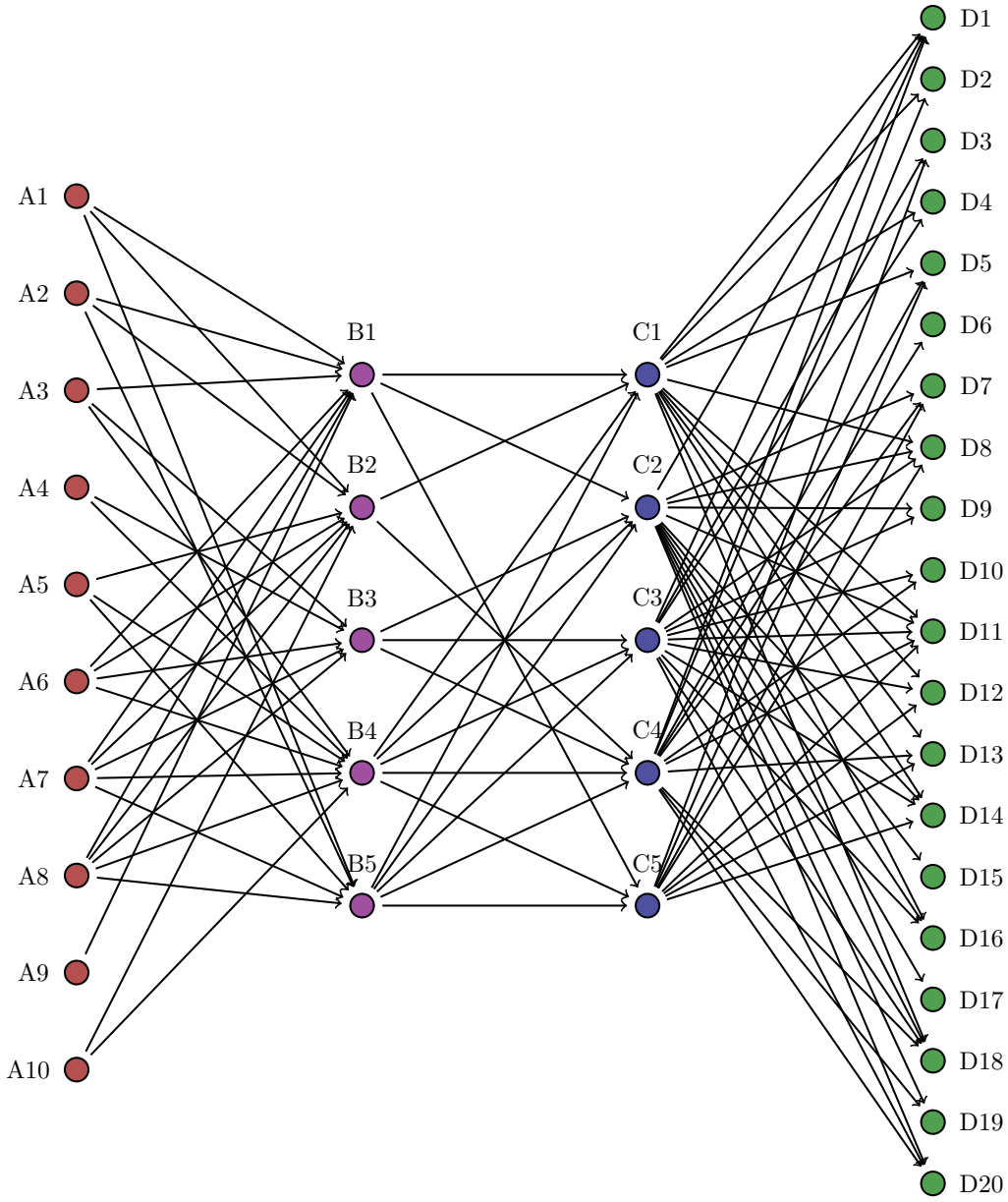


Figure 1: Visualization of Papaya Republic's supply chain network.

Column	Description
`SourceNodeName`	Name of node (cf. `NodeName` in `MCNF_nodes.csv`) from which the edge originates.
`DestinationNodeName`	Name of node (cf. `NodeName` in `MCNF_nodes.csv`) at which the edge ends.
`Capacity`	Maximum number of container units that can be transported over this edge.
`TransportationCost`	Cost incurred for each container unit transported over this edge.
`SourceNodeID`	ID (cf. `NodeID` in `MCNF_nodes.csv`) corresponding to `SourceNodeName`
`DestinationNodeID`	ID (cf. `NodeID` in `MCNF_nodes.csv`) corresponding to `DestinationNodeName`

Note that each edge in this network is directed. For example, row 3 of `MCNF_edges.csv` is the edge that goes from A1 to B5, which has a capacity of 54 units and a transportation cost of 5. This means that at most 54 container units can be transported from A1 to B5, and each container unit costs 5 to transport. Note that container units cannot be transported backwards, i.e., we cannot transport container units from B5 to A1.

Lastly, all costs are measured in cost units, i.e., a `TransportationCost` of 5 means each container unit results in 5 cost units.

Part 1: Understanding the data

Before formulating our problem, we will first understand the two data sets.

a) How many nodes are in the network?

b) How many edges are in this network?

c) What is the total amount of inventory over all nodes in layer A? What is the total amount of inventory over all nodes (all four layers)?

d) What is the total demand?

Part 2: Determining an optimal transportation plan

In this part of the problem, we wish to determine how many units Papaya Republic should transport through this supply chain network. The firm needs to determine how many container units to transport along each edge. It needs to so as to (1) not exceed the capacity of each edge and (2) so as to meet the demand in the nodes in layer D. The goal of the firm is to minimize its total transportation cost.

a) Formulate this minimum cost network flow problem as a linear program.

Hint: in this network flow problem, supply and demand are not balanced, i.e., the total supply is greater than the total demand. Thus, when you formulate the flow constraint for each node, your constraint should capture the following logic:

$$\text{Total flow into } i + \text{Inventory at } i \geq \text{Total flow out of } i + \text{Demand at } i \tag{1}$$

i.e., any outflows from a node plus the demand that needs to be met, should not be more than the total inflows plus any available inventory already at the node.

b) Implement and solve your linear program. What is the optimal cost in cost units?

c) How many units are transported along the edge from C4 to D18?

Part 3: Expanding and reorganizing the supply chain network

In this part of the problem, Papaya Republic is interested in understanding how it might re-organize or expand its supply chain network to reduce the overall transportation cost. Each of the questions below looks at several different proposals for how we might adjust the network to improve the overall cost.

a) Suppose that we are able to expand the capacity along any single edge by 20 units. Which edge would you choose for capacity expansion, and why? By how much would the overall cost of the transportation plan improve?

b) Suppose that we are able to increase the inventory at any of the nodes in layer A by a total of up to 50 units. Which node(s) would you choose to increase the inventory at, and by how much, so as to most reduce the overall cost of the transportation plan?

c) Suppose that we are able to add a new edge between any two nodes in adjacent layers, for which there does not exist an edge. (For example, we could add an edge from A1 to B3, as there is no edge between those two nodes. However, we cannot add an edge from A1 to C2, because these two nodes are not in adjacent layers.) The new edge would have a capacity of 50 units, and a unit cost of 0 (i.e., there is no cost for transporting any number of container units over this edge). Which edge would you add to most reduce the overall cost of the transportation plan?

Part 4: Simulating Disruption Scenarios

In this part of the question, we will consider several scenarios to understand how the supply chain will respond to disruptions. For this part of the question, we will modify the model from Part 2 in the following way. Unlike the model in Part 2, we now allow for the demand at each of the nodes in the D layer to only be partially satisfied. However, any shortfall accrues an additional *backorder cost* to the firm. The column `BackorderCost` in `MCNF_nodes.csv` contains the unit backorder cost for each node in the supply chain. Note that this column contains zeros for all nodes outside of layer D, but contains non-zero values for the nodes in layer D.

a) Formulate a modified version of your LP in Part 2 that minimizes the total of the transportation and backorder costs.

b) Implement your LP and solve it. What is the optimal total cost (sum of transportation and backorder costs)?

We will refer to the optimal total cost in (b) as the optimal cost of the *base scenario*, i.e., where the available inventories, the edge capacities, the edges and the transportation costs are as given in the data files.

Let's now consider two scenarios for how the supply chain network might be disrupted.

c) Suppose that we reduce the available inventory at every node by $p \times 100\%$. For example, if $p = 0.05$, and the nominal inventory at node i is S_i, then node i's inventory would be reduced by 5%, i.e., it would now have $(1 - p) \times S_i = 0.95S_i$ container units of inventory.

For each value of $p \in \{0.01, 0.02, \ldots, 0.15\}$, solve your LP from (a) and record the optimal cost. Plot your optimal cost as a function of p. By what percentage does the optimal cost increase if inventory is decreased by 10% (i.e., $p = 0.10$)?

d) Consider the following disruption scenario: 5 of the edges are chosen uniformly at random from the supply chain network and are removed. Perform 100 simulations of this type of disruption, where in each simulation you solve your LP from (a) and record the optimal cost. (Set your random seed to 100 beforehand.) What is the average cost over the 100 simulations? How much does the mean overall cost increase relative to the base case?

Hint: This question involves resolving the LP from (a) / (b). If you formulated your decision variables that represent the flow along an edge using the `ub = ...` optional parameter in the Gurobi `addVar` / `addVars` function, you can set the capacity of the edge to zero by writing

```
x[t].ub = 0
```

where `t` is the index of an edge. This has the same effect as removing that edge from the network. If you formulated it using a constraint, e.g.,

```
flow_constr[t] = m.addConstr(x[t] <= capacity[t])
```

then you can write

```
flow_constr[t].rhs = 0
```

to set that edge's capacity to zero. After you update the capacities, do not forget to update the model, i.e., run `m.update()`. (Also, if you are modifying the same model for each simulation, do not forget to restore the capacities to the edges after each simulation!)

2 Cargo freight scheduling

A freight company operating out of Los Angeles, CA needs to make a number of deliveries by air to a set of cities in the continental United States. Each city corresponds to one delivery.

The company has a single DC-10-30 aircraft available to make these deliveries, and needs to decide in what order to make deliveries (i.e., in what order to visit these cities) so as to minimize the time traveled. The aircraft departs from Los Angeles, and must return to Los Angeles after visiting all of the remaining 67 cities. Each city must be visited exactly once.

You are provided a data set (`cargo-city-locations.csv`) consisting of the locations of the cities by their latitude and longitude coordinates. There are 68 cities, including Los Angeles.

Part 1: Building our data

In order to solve the problem, we will need to calculate the flight time between each pair of cities in the data set. We will do this by applying a little bit of geometry.

For each pair of cities i and j, first convert the latitudes LAT_i and longitudes LON_i, which are given in degrees, to radians:

$$\begin{aligned}
\text{LAT}_{r,i} &= \text{LAT}_i/360 \times 2\pi, \\
\text{LON}_{r,i} &= \text{LON}_i/360 \times 2\pi, \\
\text{LAT}_{r,j} &= \text{LAT}_j/360 \times 2\pi, \\
\text{LON}_{r,j} &= \text{LON}_j/360 \times 2\pi.
\end{aligned}$$

Then, we will use the *haversine formula* to calculate the angle corresponding to the great circle distance (see Figure 2 below) between cities i and j. The *haversine* $H_{i,j}$ of cities i and j can be calculated as

$$H_{i,j} = \frac{1 - \cos(\text{LAT}_{r,j} - \text{LAT}_{r,i})}{2} + \cos(\text{LAT}_{r,i}) \times \cos(\text{LAT}_{r,j}) \times \frac{1 - \cos(\text{LON}_{r,j} - \text{LON}_{r,i})}{2} \tag{2}$$

We can now calculate the distance, $d_{i,j}$ as

$$d_{i,j} = 2r \sin^{-1}(\sqrt{H_{i,j}}) \tag{3}$$

where r is the radius of the earth, which we can assume to be 6378.137 km.

Lastly, you can use the fact that the average cruise speed of a DC-10 aircraft is 908 km / hour to convert these distances into travel times. This will yield us an approximate travel time between each pair of cities.

To help you with these calculations you can use the `numpy` functions `cos`, `sin`, `arcsin` (the arcsine function, which is just the inverse sine function $\sin^{-1}(\cdot)$) and `sqrt` (the square root).

To verify that you have implemented your distance calculations correctly, you should find that the travel time from Des Moines, IA to Baton Rouge, LA is 1.3857 hours, corresponding to a distance of 1258.226 kilometers.

a) Which two cities have the highest travel travel time?

b) Which two cities have the smallest travel time?

c) Which city has the smallest average travel time to all of the other cities? (Here, the average is taken over the other 67 cities.)

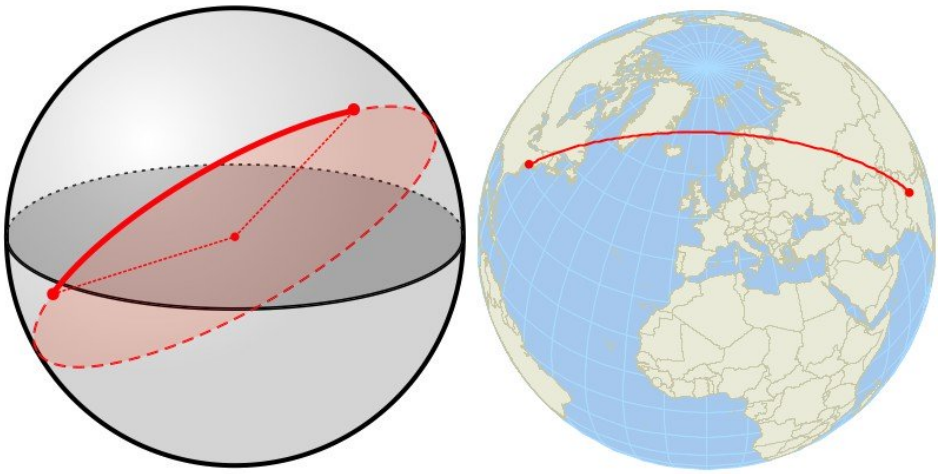


Figure 2: Visualization of great circles (grey and red circles on the left) and the great circle distance between New York City and New Delhi (right). (Source: `https://www.caliper.com/glossary/what-is-a-great-circle.htm`).

Part 2: Finding a schedule

a) Suppose that we randomly selected a sequence of cities. In `numpy`, you can construct a random sequence by using the commands

```
import numpy as np
nCities = 68
temp = np.random.permutation(nCities)
```

which will generate a list with the integers from 0 to 67 in random order. You can then calculate the travel time by iterating through the list and looking up the travel time of consecutive cities in the list, and adding them up. At the end, you would finally add the travel time corresponding to traveling from city `temp[67]` to city `temp[0]`.

Set your seed to 50. Randomly generate 100 sequences of the 68 cities, and calculate the total travel time required to visit the cities according to each sequence. What is the average of the total travel times of these 100 randomly generated sequences, in hours?

b) Suppose that we design the sequence of cities using the following heuristic. Starting from Los Angeles, the next city in the schedule is the one that is closest to the current city in travel time and has not been visited yet. What is the total travel time of this sequence, in hours?

c) Solve an optimization problem to find the order in which the cities should be visited, so as to minimize the total travel time. What is the total travel time of this sequence, in hours?